

- 18    WHYTE, R.J. & GREEN, M.E.    Geology and Palaeogeography of Chibuluma West Orebody Zambian Copperbelt - Economic Geology 66, pp. 400-424.    1971.
  
- 19    WHITE, W.S. & WRIGHT, J.C.    The White Pine Copper Deposit, Ontonagon County, Michigan, Economic Geology 49, pp. 675-716.    1954.
  
- 20    STANTON, R.L.    General Features of Conformable 'Pyritic' Orebodies.    Part I, Field Association. Trans. Canadian Inst. of Mining & Metallurgy 63, 1960, pp. 22-27. (Canadian Min. Metall. Bull. No. 573, January 1960, pp. 24-29)
  
- 21    DARNLEY, A.G.    Petrology of some Rhodesian Copperbelt orebodies and associated rocks.    Discussion by P. Gilmour. Trans. Inst. Min. Metall. London 69. 1959-1960. (Bull. Inst. Min. Metall. London No. 646, 1960) pp. 699-700.
  
- 22    FRASER, D.C.    Organic Sequestration of Copper - Economic Geology 56, pp. 1063-1078, 1961.
  
- 23    OBERC, J. & SERKIES, J.    Evolution of the Fore-Sudetic Copper Deposit - Economic Geology 63, pp. 372-379, 1968.
  
- 24    BROWN, J.S.    South-east Missouri Lead Belt, Field Trip No. 1., Geological Society America Guide Book for Field Trips - 1958.
  
- 25    GRIP, E.    The Lead Deposits of the Eastern Border of the Caledonides in Sweden.    Proc. Int. Geol. Congr., 21st Session, 1960, pp. 149-159.

- 26 GIGNOUX, M. Stratigraphic Geology (trans. by G.G. Woodford)  
-published by W.H. Freeman & Co., San Francisco - 1955.  
p. 215.
- 27 BURGER, D. Private communication.
- 28 ROEDDER, E. Fluid inclusions as samples of the ore  
forming fluids. Berlingske Bogtrylklein XXI  
International Geological Congress Procs. of  
Section 16 - 'Genetic problems of ores' 1959  
pp. 218-229.
- 29 ROEDDER, E., INGRAM, B. & HALL, W.F. Studies of fluid  
inclusions III : Extraction and Quantitative  
Analysis of inclusions in the milligram range.  
Economic Geology 58, 1963, pp. 353-374.

P.J. CURTIS  
(EMH)

**PACMINEX PTY. LIMITED**

000147

MEMORANDUM TO: DR. K. MAIDEN

OUR REF: PJC/IM

FROM: P.J. CURTIS

DATE: 15.10.1973

ROCKS FOR PETROLOGICAL -  
MINERALOGICAL EXAMINATION

INTRODUCTION

Corraberra Sandstone samples from Mt. Gunson; pyrite rich ore from Shaft 2, Cattle Grid; possible Tregolana Shale from Elizabeth Creek; and, possible Whyalla Sandstone from Elizabeth Creek were submitted for petrographic examination.

SUMMARY

A 9855  
(70736)

The rock consists of mainly originally round, now much silicified quartz grains with authigenic overgrowths. The rock also contains a little chert, appreciable microcline, octahedra as limonite: leucosene pseudomorphs after magnetite and a little interstitial clay mineral. Bedding is shown by parallel bands in the rock where 'magnetite' is concentrated.

"Corraberra Sandstone"

A 9856  
(70737)

As for A 9855 but with closer spaced magnetite bands.

"Corraberra Ss"

A 9857  
(70738)

Similar to A 9855 but magnetite is scattered through the section.

"Corraberra Ss"

A 9858  
(70739)

The sample consists of acicular prisms of gypsum and traces of limonite impurity.

A 9859  
(70740)

Original syngenetic pyrite precedes bornite: chalcopryrite mineralisation. Much of the later pyrite which has precipitated around the syngenetic pyrite may be slightly earlier or contemporary with bornite: chalcopryrite intergrowths.

"Cattle Grid ore"

A 9860  
(70741)

Pyrite is primary where it occurs here. The typical, carrollite → chalcopryrite → bornite chalcocite → sphalerite with galena paragenesis is seen. The carrollite is mainly a nucleating material for following secondary copper sulphides and was not

"Cattle Grid ore"

appreciably involved in the replacement steps. Pyrite has left few relicts but is seen mainly as solitary grains in the quartzite interstices; however there is occasional evidence of replacement by sphalerite.

A 9861

Tregolana Shale similar to this material has not been observed by the author. Some round quartz grains, as seen in Whyalla Sandstone are observed but most grains are large consisting of quartz rock and are also somewhat angular. Other Whyalla Sandstone characteristics (ferruginous cherts, etc) are also rarely seen. It is thought the deposit is more recent, with some derived grains of Whyalla Sandstone.

A 9862

Whyalla Sandstone with large round quartz grains, medium sized ferruginous cherts, silt-sized quartz particles with a large size interval between silt size and sand size median grains. The remaining matrix is of fine limonitic quartz.

#### DETAILS OF INVESTIGATION

A 9855  
(70736)

An even grained sandstone of quartz grains (.05 → .24mm, median .16mm) of angular to rounded grains, mainly the latter. Patches of quartz mosaics of silicified quartzite show original outlines of more rounded grains as internal ellipitical trains of fluid inclusions and opaque particles. Between the silicified areas are areas where vugs of clay mineral (probably illite > kaolinite) occur and where quartz grains show no authigenic overgrowths. Fresh, twinned microcline grains are seen of similar size to neighbouring quartz grains but are often rectangular shaped with rounded corners.

Rounded cherts with some admixed clay mineral and of similar size to quartz grains are scattered around the section. Occasional detrital muscovite flakes, somewhat bent (median .3 x .05mm) are present interstitially. Limonite: leucoxene masses pseudomorphing magnetite euhedra (.03 to .10mm, median .06mm) occur scattered throughout and interstitially. At intervals, the magnetite pseudomorphs are more abundant in narrow horizons along the 'bedding' plans of the rock where otherwise the bedding is not immediately obvious. A proportion of fine detrital hematite after magnetite particles are seen in the interstices, particularly with clay mineral. A little rounded brown tourmaline is seen (~.08 x .04 mm), a little topaz of similar grain size, and

also some detrital zircon grains.

Modal mineral composition: -

| <u>Mineral</u>                        | <u>Volume percent</u> |
|---------------------------------------|-----------------------|
| Quartz                                | 89                    |
| Chert                                 | 2                     |
| Microcline                            | 3                     |
| Muscovite                             | < 1                   |
| Clay Mineral                          | 2                     |
| Limonite/leucoxene<br>after magnetite | 3                     |
| Tourmaline topaz,<br>zircon           | < 1                   |

The rock is somewhat similar to Corraberra Sandstone (A 4224) described in PMR Report 195/72 (Ref. 2).

A 9856  
(70737)

Very similar to sample A 9855 in quartz grain sizes, shapes and the degree of silicification. In hand specimen the rock is apparently less massive and more flaggy than is A 9855. This is emphasised in thin-section by repeated parallel, inequispaced intervals (1 to 3mm) of magnetite euhedra (limonite: leucoxene pseudomorphs), as 0.1 to 0.3mm width concentrates associated with the normal sandstone fabric. Except for the close formed 'magnetite' bands the rock is texturally similar to A 9855.

Modal mineral composition: -

| <u>Mineral</u>                                               | <u>Volume percent</u> |
|--------------------------------------------------------------|-----------------------|
| Quartz                                                       | 86                    |
| Chert                                                        | 3                     |
| Microcline                                                   | 2                     |
| Muscovite                                                    | 1                     |
| Clay mineral                                                 | 3                     |
| Magnetite (limonite:<br>leucoxene mostly in<br>pseudomorphs) | 5                     |
| Tourmaline (blue green<br>and brown, rounded,<br>detrital)   | < 1                   |
| Zircon                                                       | << 1                  |

A 9857  
(70738)

Texturally very similar to A 9856 in grain size variation of quartz grains, silicification of the quartz, and the presence of fresh microcline grains (often showing relatively unworn oblong shapes). The presence of brownish (volcanic ?) cherts which have volcanic flow banded, glassy

textures is a difference from A 9856. Also the pseudomorphs of magnetite (limonite: leucoxene aggregates) grains are more scattered through the section. Detrital tourmaline present is greenish-brown, pleochroic and very worn.

Modal mineral composition: -

| <u>Mineral</u>                     | <u>Volume percent</u> |
|------------------------------------|-----------------------|
| Quartz                             | 89                    |
| Chert                              | 2                     |
| Microcline                         | 3                     |
| Muscovite                          | < 1                   |
| Clay Mineral                       | 2                     |
| Magnetite (limonite:<br>leucoxene) | 3                     |
| Tourmaline > zircon                | 1                     |

A 9858  
(70739)

Material filling fractures in Corruberra Sandstone - probably gypsum.

Fine, transparent, laths forming an unoriented aggregate (as seen under the low power stereo-microscope).

Immersed in R.I. oils the laths are seen to be of variable width (5 to 15  $\mu$ ) and of variable length (to 0.3mm); occasionally squat prisms to 50  $\mu$  wide with terminal crystal forms are seen.

Refractive indices  $n_{\alpha} = 1.520$ ,  $n_{\beta} = 1.524$   
and  $n_{\gamma} = 1.529$ .

A few crystals of sample were dissolved in dilute HCL solution and a drop of 10% barium chloride solution was added. A copious, white, crystalline precipitate of  $Ba SO_4$  indicated the presence of sulphate.

The sample is of the selenite variety of gypsum.

A 9859  
(70740)

Sample of pyrite-rich ore from Shaft 2.

Possible second generation?

Cubes and pyritohedra of pyrite mainly about 0.13mm in grain size are seen to be precipitated around earlier pyritohedra and occasional octahedra of greigite (?). Blades of marcasite are seen as outgrowths of the pyrite or occur as solitary grains of pseudomorphed pyrite. Sphalerite is seen to be replacing pyrite

locally but where more massive it is filling voids in the rock. In one place sphalerite has mutual boundaries with marcasite. Lepidocrocite laths are seen as intergrowths with pyrite. Studying the Eh/pH stability diagram of iron sulphides and oxides in water at 25°C and 1 atmosphere total pressure and total dissolved sulphur =  $10^{-1}$ , (Garrels and Christ, Ref. 1), it is seen that pyrite and ferric iron oxides will precipitate together at acid to neutral pH and a slight positive Eh. Where pyrite and copper sulphides (bornite: digenite: chalcopyrite, intergrowths) occur together some local replacement of pyrite is seen (Figure 1). On the other hand there are seen to be more mutual boundaries between pyrite and the copper sulphides. This would suggest that precipitation of some pyrite occurred just before and simultaneously with the copper sulphides.

The nature of pyrite precipitation around preformed pyrite nuclei indicates the continuing precipitation of the mineral during diagenesis. The original pyrite nuclei were probably formed from gel, iron sulphide or hydroxide in the presence of sulphide swept in with the clay during rock deposition.

The thin section shows an abundance of clay-mineral, partly fibrous and part as kaolinite platelets. Narrow channels of clay mineral occur between many of the quartz grains but groups of the grains are silicified into mosaics. Most pyrite occurs in channels and vugs where clay occurs. A small amount of the sulphide occurs trapped by authigenic quartz overgrowths. Euhedral forms prevail showing that pyrite growth occurred prior to silicification and consequent isolation from percolating solutions.

A 9860  
(70741)

As for A 9859, pyrite mineralisation.

Polished section of the ore shows bornite being replaced by chalcocite which is in turn being replaced by sphalerite. Relict carrollite, the main original sulphide nuclei occur as octahedra or as embayed forms within the chalcocite. A little late galena is replacing chalcocite at the boundaries between carrollite and chalcocite. The carrollite contains inclusions of exsolved chalcocite (mutual boundaries) and along lattice plane fractures. Some very fine bornite: chalcopyrite intergrowth has exsolved along chalcocite lattice planes, (forms later than the already replaced bornite within the chalcocite mass). This

indicates a very slight paragenesis reversal due to locally changed physical conditions.

The minor pyrite content seen here occurs mainly away from other sulphides embedded in the quartz gangue, although it shows continuing precipitation of later stages of pyrite around earlier pyrite nuclei. When the pyrite does occur sparsely as inclusions in sulphide masses, it shows replacement (embayment) by sphalerite.

The main primary sulphide here is carrollite. Pyrite occurs associated only with sphalerite of all the sulphides present; otherwise it is shown to be earlier formed.

A 9861  
(70742)

Sandy clay from Elizabeth Creek. Possible Tregolana Shale?

This specimen does not look like any Tregolana Shale sections that has been seen so far.

Larger grains are equally represented by quartz mineral grains and quartz rock fragments (.03 to 1mm, median ~ .48mm). A few (< 20 percent large grains) which are quite rounded as seen in the plane of thin-section, suggest a Whyalla Sandstone derivation for these. (Figure 2). A very small number of rounded, subangular, brown cherty grains which are present also suggest an origin from Whyalla Sandstone. Remaining large quartz grains could well have originated from Pandurra Quartzite, since all are to some degree angular.

The matrix, about 40 volume percent of rock volume, shows part silicification by secondary chert. Replacement of grain margins of some of the large quartz grains by clay mineral makes the overall texture a little obscure. Derivation of larger round quartz grains from a Whyalla Sandstone similar to A 9842 (70726) West Lagoon is possible.

Most likely a relatively more recent (Tertiary to Recent?) deposit as suggested (Ref. 3), with derived Whyalla Sandstone material mixed with rock material from other underlying or adjoining formations.

A 9862  
(70743)

A ferruginous deposit from Elizabeth Creek.

A ferruginous Whyalla Sandstone or a type of ferricrete?

In hand specimens the rock has all the

appearances of a residual ferricrete but in thin section there can be no mistake about its identity. All large grains (0.2 to 4.0mm, median 0.5mm) are very round, (Figure 3). About 20 volume percent of the grains are quartz rock fragments and the remainder are quartz mineral grains. As would be expected the quartz rock fragments are mainly the larger grains. The typical rounded ferruginous cherts as seen in Whyalla Sandstone are also present to at least 5 volume percent of the large grains present. Some of the latter (~ 0.2mm) are partly masked by the ferruginous matrix of the rock. The matrix is partly of roughly silt size subangular quartz grains (.016 to .080mm, median 0.05mm) and the clear quartz grains constitute as far as can be seen, about 10 volume percent of the matrix. The remaining matrix is of ferruginous silt size quartz and limonite. Limonite tends to encroach on the margins of quartz grains.

#### REFERENCES

1. Garrels R.M. and Christ, C.L. - 1965  
Solutions, Minerals and Equilibrium  
pb. Harper and Row, N.Y. London
2. Curtis, P.J.  
Petrography of Mt. Gunson Sediments  
PMR 195/72.
3. Memorandum from K. Maiden to P.J. Curtis;  
Rocks for Petrological - Mineralogical  
examination (dated 27th August, 1973).

*P.J. Curtis*  
.....

**PACMINEX PTY. LIMITED**

MEMORANDUM TO: Mr. P. Curtis

OUR REF: KJM/MM

FROM: K. Maiden

DATE: 27th August, 1973

ROCKS FOR PETROLOGICAL - MINERALOGICAL EXAMINATION

Would you please examine the following samples:

1. Samples 70736-38: Samples of Corraberra Sandstone from Mt. Gunson; general petrographic description required.
2. Sample 70739: Material filling fractures in Corraberra Sandstone; possible gypsum.
3. Samples 70740-41: Samples of pyrite-rich ore from shaft 2. Note that, in 70740 the pyrite occurs in a zone parallel to bedding. Is this a second generation of pyrite?  
i.e. formed after the first generation pyrite was replaced by copper sulphides?
4. Sample 70742: Sandy clay from Elizabeth Creek. Jim Heape considers this to be Tregolana Shale. Other possibilities are:-
  - (i) Dolomitic clay (Woocalla Dolomite equivalent)
  - (ii) Tertiary to Recent deposit
  - (iii) Clayey facies of Whyalla Sandstone
5. Sample 70743: Ferruginous deposit from Elizabeth Creek. Is this a ferruginous Whyalla Sandstone, or a type of ferricrete.

K.J. Maiden (Signed)  
23.8.73 - Mt. Gunson.

c.c. Mr. R. Gersteling



Figure 1  
Pyrite-rich ore from Shaft 2, Cattle Grid (70740/A9859).

Finely intergrown bornite (light-grey) and chalcopyrite (off-white) comprise most of the sulphide mass (mainly white area). Pyrite euhedra (white) crop along the north, north-east, south and south-west edges of the sulphide mass. Some incursion of pyrite by copper sulphides is seen but mainly straight edge mutual boundaries prevail.

Plain light

Magnification X 300

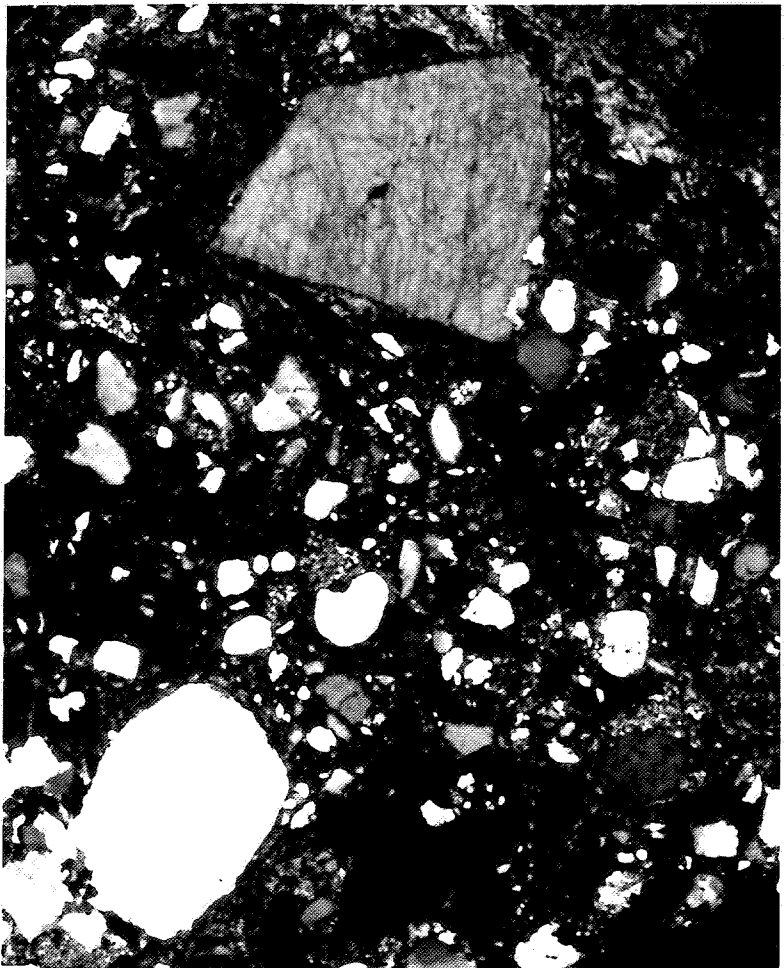


Figure 2

Sandy clay from Elizabeth Creek thought to be Tregolana shale but more likely a much later deposit. (70742/A9861).

Many rounded quartz grains and few small ferruginous cherts (speckled black and white) are probably derived from Whyalla Sandstone. More angular grains may have derived from pandurra Quartzite. Rock matrix is mainly of clay mineral (grey and white stipple); secondary chert (dark areas), replacing clay. Scattered, fine quartz grains (white) are also seen.

Crossed nicols

Magnification X 15

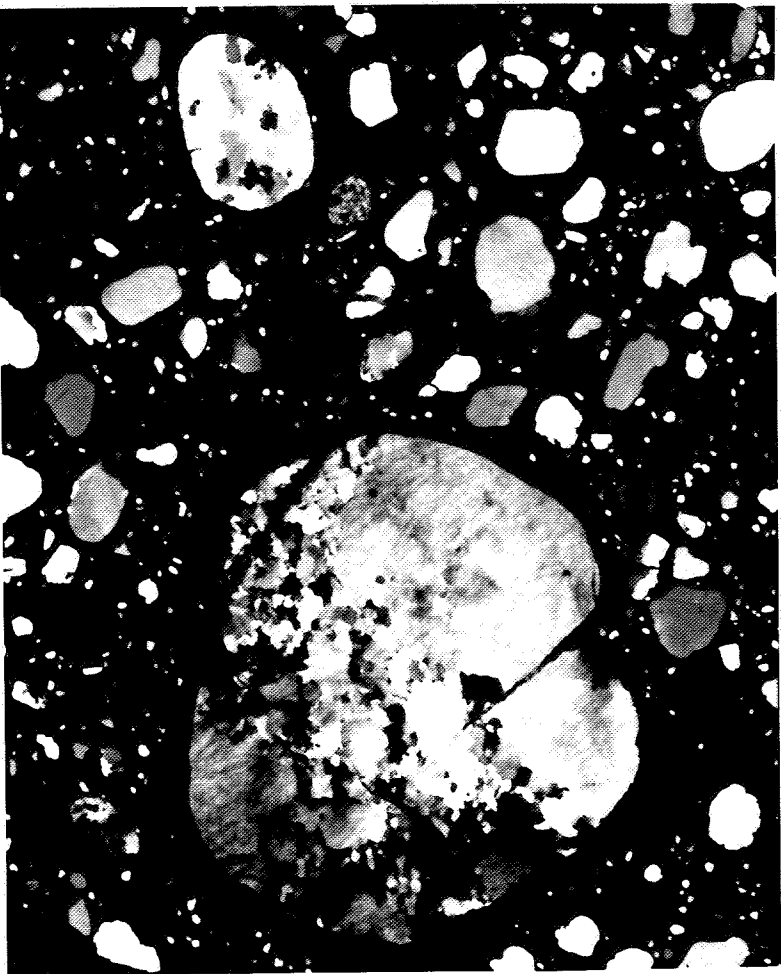


Figure 3

Ferruginous deposit from Elizabeth Creek (70743/A9862).

Large round quartz rock grains (grey and white mosaic patterns) and quartz mineral grains (flat white and grey) are seen. Ferruginous cherts (speckled black, white and grey) are present. Several rounded cherts are obscured by iron staining. The groundmass is largely of fine limonitic quartz (black) and fine clear silt size quartz grains (white). Whyalla Sandstone.

Crossed nicols

Magnification X 15